CS 2613 QUIZ 1

Thursday, 19 February 2026 11:24 pm

$F(A,B,C) = \sum m(0,1,2,6,7)$

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

A \ BC	00	01	11	10
0	1	1	0	1
1	0	0	1	1

$F = \bar{A}\bar{B} + AB + BC'$

$F(A,B,C) = \sum m(0,2,3,4,6)$

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1

A \ BC	00	01	11	10
0	1	0	1	1
1	1	0	0	1

0 & 4

2 & 6

3

$F = \bar{A} + \bar{B}\bar{C}$

$F = A'BC + C'$

$F(A,B,C) = \sum m(1,2,3,5,7)$

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

A \ BC	00	01	11	10
0	0	1	1	1
1	0	1	1	0

1 & 3

5 & 7

2

$F = A'B + BC + A'BC'$

$\boxed{F = A'B + BC}$

$F(A,B,C) = \sum(0,1,3,4,5,7)$

A \ BC	00	01	11	10

0 & 1

$F(A,B,C) = \Sigma(0,1,3,4,5,7)$

A \ BC	00	01	11	10
0	1	1	1	0
1	1	1	1	0

0 1
4 5

$F = B'C' + B'C + A'B + AB$

$F = B' + B \quad F = 1$

$F(A,B,C) = \Sigma(0,2,3,4,6,7)$

A \ BC	00	01	11	10
0	1	0	1	1
1	1	0	1	1

$F = B'C' + B$

$F = (B' + B)(B + C')$

$F(A,B,C) = \Sigma(0,2,5) \quad d(A,B,C) = (6,7)$

A \ BC	00	01	11	10
0	1	0	0	1
1	0	1	X	X

A	B	C	
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	X
1	1	1	X

$F = A'C' + AC$

$= (A \odot C)$

$F(A,B,C,D) \Sigma(0,1,2,6,8,9,11)$

AB \ CD	00	01	11	10
00	1	1	X	1

AB \ CD	00	01	11	10
00	1	1		1
01				1
11				
10	1	1	1	

$F = B'C' + A'CD' + AB'D$

$F(A,B,C,D) = \Sigma(5,6,9,12,13)$

$d(3,4,7,10,11,14,15)$

AB \ CD	00	01	11	10
00	0	0	X	0
01	X	1	X	1
11	1	1	X	X
10	0	1	X	X

$F = B + AD$

$F(A,B) = \Sigma(0,1,3)$

A \ B	0	1
0	1	1
1		1

$F = B + A'$

2. $F(A,B,C) = \Sigma(2,3,4)$

BC	00	01	11	10

2. $F(A,B,C) = \sum(2,3,4,5)$

A \ BC	00	01	11	10
0			1	1
1	1			

$F = A'B + AB'C'$

3. $F(A,B,C) = \sum(0,1,2,4,5,7)$

A \ BC	00	01	11	10
0	1	1		1
1	1	1	1	

000
001
100
100
101
111

$F = B' + A'C' + AC$

$= AC + A'C' + B'$

4. $F(A,B,C,D) = \sum(3,4,5,6,7,12,14)$

AB \ CD	00	01	11	10
00			1	
01	1	1	1	1
11	1			1
10				

$F = A'B + BD' + A'CD$

$F = A'B + A'CD + BD'$

5. $F(A,B,C,D) = \sum(0,1,2,4,5,7,11,15)$

AB \ CD	00	01	11	10
00	1	1		1
01	1	1	1	
11			1	
10			1	

0111

$F = A'C' + ACD + A'B'D' + A'BCD$

6. $F(A,B,C,D) = \Sigma(0,1,2,3,4,6,8,9,10,11,12,14)$

AB \ CD	00	01	11	10
00	1	1	1	1
01	1			1
11	1	1		1
10	1	1		1

1100
1101
1000

$F = A'B' + D' + AC'$

7 $F(A,B,C,D) = \Sigma(5,6,9,12,13)$
$d(3,4,7,10,11,14,15)$

AB \ CD	00	01	11	10
00			X	
01	X	1	X	1
11	X	1	X	X
10		1	X	X

1101
1001
1111
1011

$F = B + A$

$= B + AD$

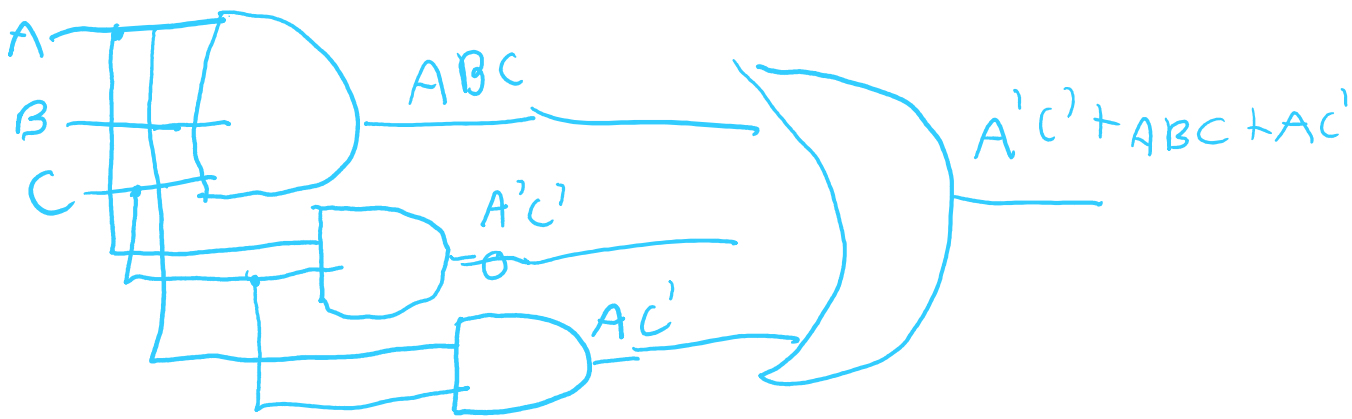
$= AD + B$

Simplify

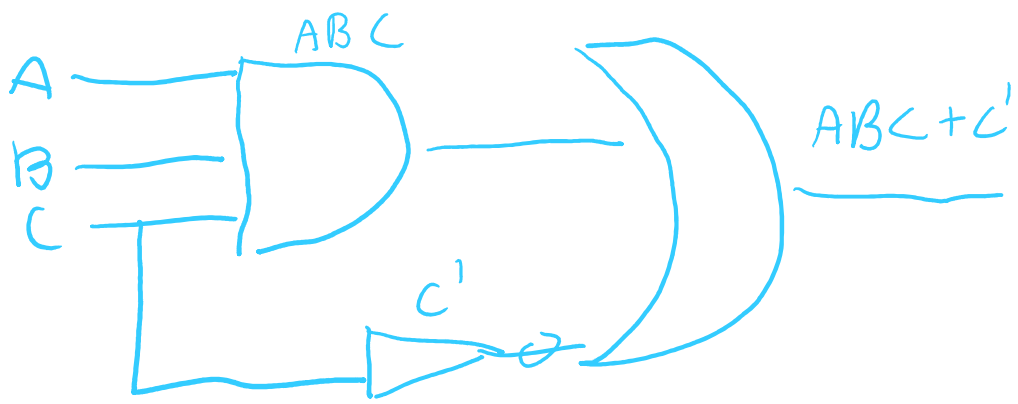
$A'C' + ABC + AC'$

$C'(\cancel{A' + A})$

$1 \quad C' + ABC$



$C' + ABC$



$1 \quad 0$

0 1

1 2

0 3

1 4

0 5

1 6

1 7

$F(A,B,C,D)\ \Sigma(0,6,8,)$

$d(2,4,10)$

AB \ CD	00	01	11	10
00	1			X
01	X			1
11				
10	1			X

0 0 0 0

0 0 1 0

0 1 0 0

0 1 1 0

$F = B'D' + A'D'$